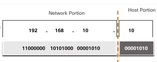
IPv4 Address Structure

TALK

Network and Host Portions

An IPv4 address is a 32-bit hierarchical address that is made up of a network portion and a host portion. When determining the network portion versus the host portion, you must look at the 32-bit stream, as shown in the figure.

IPv4 Address



The bits within the network portion of the address must be identical for all devices that reside in the same network. The bits within the host portion of the address must be unique to identify a specific host within a network. If two hosts have the same bit pattern in the specified network portion of the 32-bit stream, these two hosts will reside in the same network.

But how do hosts know which portion of the 32-bits identifies the network and which identifies the host? That is the role of the subnet mask.

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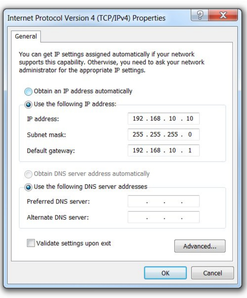
The Subnet Mask

As shown in the figure, assigning an IPv4 address to a host requires the following:

- **IPv4 address** – This is the unique IPv4 address of the host.
- **Subnet mask** – This is used to identify the network/host portion of the IPv4 address.

IPv4 Configuration on a Windows Computer



Note: A default gateway IPv4 address is required to reach remote networks and DNS server IPv4 addresses are required to translate domain names to IPv4 addresses.

The IPv4 subnet mask is used to differentiate the network portion from the host portion of an IPv4 address. When an IPv4 address is assigned to a device, the subnet mask is used to determine the network address of the device. The network address represents all the devices on the same network.

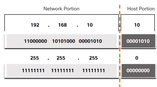
The next figure displays the 32-bit subnet mask in dotted decimal and binary formats.

Subnet Mask

Notice how the subnet mask is a consecutive sequence of 1-bits followed by a consecutive sequence of 0-bits.

To identify the network and host portions of an IPv4 address, the subnet mask is compared to the IPv4 address bit for bit, from left to right as shown in the figure.

Associating an IPv4 Address with its Subnet Mask



Note that the subnet mask does not actually contain the network or host portion of an IPv4 address, it just tells the computer where to look for the part of the IPv4 address that is the network portion and which part is the host portion.

The actual process used to identify the network portion and host portion is called ANDing.

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The Prefix Length

Expressing network addresses and host addresses with the dotted decimal subnet mask address can become cumbersome. Fortunately, there is an alternative method of identifying a subnet mask, a method called the prefix length.

The prefix length is the number of bits set to 1 in the subnet mask. It is written in "slash notation", which is noted by a forward slash (/) followed by the number of bits set to 1. Therefore, count the number of bits in the subnet mask and prepend it with a slash.

Refer to the table for examples. The first column lists various subnet masks that can be used with a host address. The second column displays the converted 32-bit binary address. The last column displays the resulting prefix length.

Comparing the Subnet Mask and Prefix Length

Subnet Mask	32-bit Address	Prefix Length
255.0.0.0	11111111. 00000000. 00000000. 00000000	/8
255.255.0.0	11111111. 11111111. 00000000. 00000000	/16
255.255.255.0	11111111. 11111111. 11111111. 00000000	/24
255.255.255.128	11111111. 11111111. 11111111. 10000000	/25
255.255.255.192	11111111. 11111111. 11111111. 11000000	/26
255.255.255.224	11111111. 11111111. 11111111. 11100000	/27
255.255.255.240	11111111. 11111111. 11111111. 11110000	/28
255.255.255.248	11111111. 11111111. 11111111. 11111000	/29
255.255.255.252	11111111. 11111111. 11111111. 11111100	/30

Note: A network address is also referred to as a prefix or network prefix. Therefore, the prefix length is the number of 1-bits in the subnet mask.

When representing an IPv4 address using a prefix length, the IPv4 address is written followed by the prefix length with no spaces. For example, 192.168.10.10 255.255.255.0 would be written as 192.168.10.10/24. Using various types of prefix lengths will be discussed later. For now, the focus will be on the /24 (i.e. 255.255.255.0) prefix.

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Determining the Network: Logical AND

A logical AND is one of three Boolean operations used in Boolean or digital logic. The other two are OR and NOT. The AND operation is used in determining the network address.

Logical AND is the comparison of two bits that produce the results shown below. Note how only a 1 AND 1 produces a 1. Any other combination results in a 0.

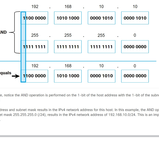
- 1 AND 1 = 1
- 1 AND 0 = 0
- 0 AND 1 = 0
- 0 AND 0 = 0

Note: In digital logic, 1 represents True and 0 represents False. When using an AND operation, both input values must be True (1) for the result to be True (1).

To identify the network address of an IPv4 host, the IPv4 address is logically ANDed, bit by bit, with the subnet mask. ANDing between the address and the subnet mask yields the network address.

To illustrate how AND is used to discover a network address, consider a host with IPv4 address 192.168.10.10 and subnet mask of 255.255.255.0, as shown in the figure:

- **IPv4 host address (192.168.10.10)** – The IPv4 address of the host in dotted decimal and binary formats.
- **Subnet mask (255.255.255.0)** – The subnet mask of the host in dotted decimal and binary formats.
- **Network address (192.168.10.0)** – The logical AND operation between the IPv4 address and subnet mask results in an IPv4 network address shown in dotted decimal and binary formats.



Using the first sequence of bits as an example, notice the AND operation is performed on the 1-bit of the host address with the 1-bit of the subnet mask. This results in a 1 bit for the network address. 1 AND 1 = 1.

The AND operation between an IPv4 host address and subnet mask results in the IPv4 network address for this host. In this example, the AND operation between the host address of 192.168.10.10 and the subnet mask 255.255.255.0 (/24), results in the IPv4 network address of 192.168.10.0/24. This is an important IPv4 operation, as it tells the host what network it belongs to.

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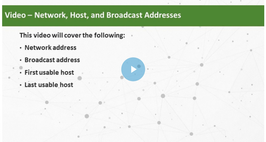
Video – Network, Host and Broadcast Addresses

Click Play to view a demonstration of how the network, host, and broadcast addresses are determined for a given IPv4 address and subnet mask.



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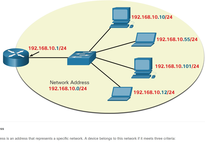
Network, Host, and Broadcast Addresses

Within each network are three types of IP addresses:

- Network address
- Host addresses
- Broadcast address

Using the topology in the figure, these three types of addresses will be examined.



Network address

A network address is an address that represents a specific network. A device belongs to this network if it meets three criteria:

- It has the same subnet mask as the network address.
- It has the same network bits as the network address, as indicated by the subnet mask.
- It is located on the same broadcast domain as other hosts with the same network address.

A host determines its network address by performing an AND operation between its IPv4 address and its subnet mask.

As shown in the table, the network address has all 0-bits in the host portion, as determined by the subnet mask. In this example, the network address is 192.168.10.0/24. A network address cannot be assigned to a device.

Network, Host, and Broadcast Addresses

	Network Portion			Host Bits	
Subnet mask **255.255.255.0 or /24**	255 11111111	255 11111111	255 11111111	0 00000000	
Network address **192.168.10.0 or /24**	192 11000000	168 10101000	10 00001010	0 00000000	All 0s
First address **192.168.10.1 or /24**	192 11000000	168 10101000	10 00001010	1 00000001	All 0s and a 1
Last address **192.168.10.254 or /24**	192 11000000	168 10101000	10 00001010	254 11111110	All 1s and a 0
Broadcast address **192.168.10.255 or /24**	192 11000000	168 10101000	10 00001010	255 11111111	All 1s

Host addresses

Host addresses are addresses that can be assigned to a device such as a host computer, laptop, smart phone, web camera, printer, router, etc. The host portion of the address is the bits indicated by 0-bits in the subnet mask. Host addresses can have any combination except for all 0-bits (this would be a network address) or all 1-bits (this would be a broadcast address).

All devices within the same network, must have the same subnet mask and the same network bits. Only the host bits will differ and must be unique.

Notice that in the table, there is a first and last host address:

- **First host address** – This first host within a network has all 0-bits with the last (right-most) bit as a 1 bit. In this example it is 192.168.10.1/24.
- **Last host address** – This last host within a network has all 1-bits with the last (right-most) bit as a 0 bit. In this example it is 192.168.10.254/24.

Any addresses between and including, 192.168.10.1/24 through 192.168.10.254/24 can be assigned to a device on the network.

Broadcast address

A broadcast address is an address that is used when it is required to reach all devices on the IPv4 network. As shown in the table, the network broadcast address has all 1-bits in the host portion, as determined by the subnet mask. In this example, the network address is 192.168.10.255/24. A broadcast address cannot be assigned to a device.

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Activity – ANDing to Determine the Network Address

Instructions:

Use the ANDing process to determine the network address (in binary and decimal formats).

Host Address	10	174	217	44
Subnet Mask	255	255	240	0
Host Address in binary	00001010	10101110	11011001	00101100
Subnet Mask in binary	11111111	11111111	11110000	00000000
Network Address in binary				
Network Address in decimal				

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Check Your Understanding – IPv4 Address Structure

Check your understanding of IPv4 address structure by choosing the correct answer to the following questions.

1. Host A has the IPv4 address and subnet mask 10.5.4.100 255.255.255.0. What is the network address of Host-A?
   - 10.0.0.0
   - 10.5.0.0
   - 10.5.4.0 (correct)
   - 10.5.4.100
2. Host A has the IPv4 address and subnet mask 172.16.4.100 255.255.0.0. What is the network address of Host-A?
   - 172.0.0.0
   - 172.16.0.0 (correct)
   - 172.16.4.0
   - 172.16.4.100
3. Host A has the IPv4 address and subnet mask 10.5.4.100 255.255.255.0. Which of the following IPv4 addresses would be on the same network as Host-A? (Choose all that apply)
   - 10.5.4.1 (correct)
   - 10.5.0.1
   - 10.5.4.98 (correct)
   - 10.0.0.98
   - 10.5.100.4
4. Host A has the IPv4 address and subnet mask 172.16.4.100 255.255.0.0. Which of the following IPv4 addresses would be on the same network as Host-A? (Choose all that apply)
   - 172.16.4.99 (correct)
   - 172.16.0.1 (correct)
   - 172.16.4.98 (correct)
   - 172.16.4.1
5. Host A has the IPv4 address and subnet mask 192.168.1.50 255.255.255.0. Which of the following IPv4 addresses would be on the same network as Host-A? (Choose all that apply)
   - 192.168.0.1
   - 192.168.4.100
   - 192.168.1.1 (correct)
   - 192.168.1.100 (correct)
   - 192.168.2.1

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